Meeting 1 of 7 · September 15, 2026

Retrospective reconstruction · prepared October 2026

RESEARCH PROPOSAL

When is quantum error mitigation worth its cost?

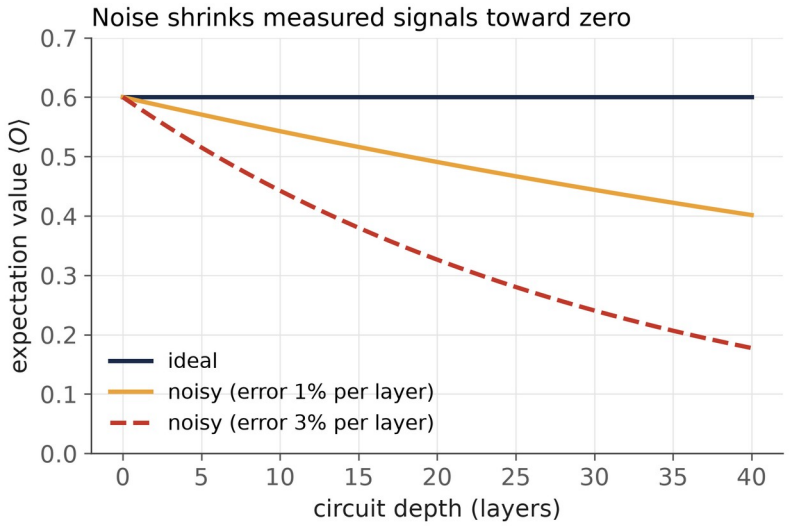
Research proposal: noise in quantum machine learning, the cost of mitigation, and a cost-aware decision layer

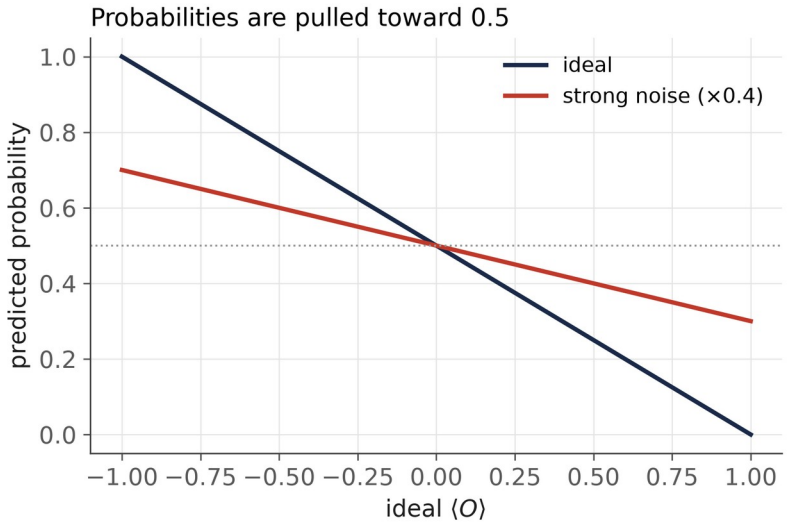
Reconstructed in Oct 2026 from the project record, which begins Oct 6. It preserves the pre-October project as a generic QML study (Iris, QM9, WDBC) with a mitigate-or-not decision. No results existed on this date.

Mayank Kumar Pokhriyal · Tushar Poonia
M.S. in Artificial Intelligence capstone · Katz School of Science and Health, Yeshiva University
Retrospective deck · proposed split: Mayank (slides 1–3) · Tushar (slides 4–5) · about 5 minutes



Why quantum noise matters





Illustrative global-depolarizing model, not project data.

In brief

- NISQ processors are noisy
- Noise shrinks the expectation values a QML model outputs
- Probabilities drift toward 0.5, so threshold decisions can flip

Noise changes what a quantum model tells us, not only how the hardware behaves.

Source: Preskill (2018), Quantum 2:79. Illustrative model only; no project data.

Why error mitigation is expensive

Zero-noise extrapolation (ZNE)

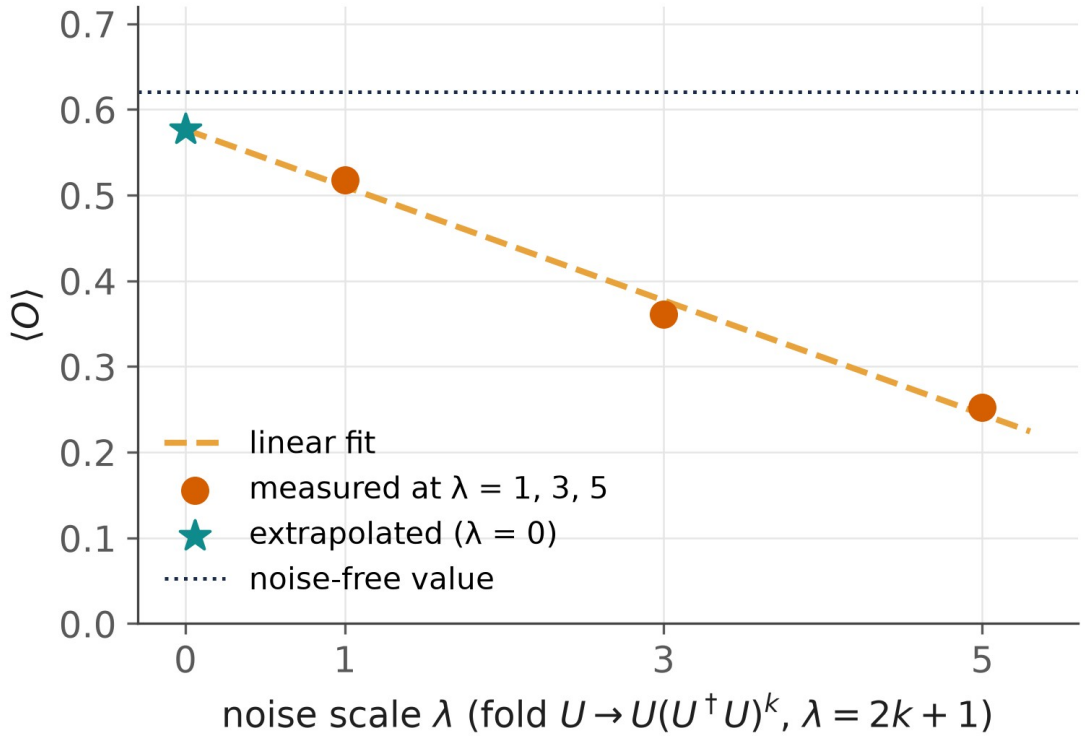
Run the circuit at several amplified noise levels, then extrapolate to zero noise (Temme et al. 2017; Kandala et al. 2019; Giurgica-Tiron et al. 2020).

The cost

ZNE: several circuits instead of one. Readout mitigation: a few calibration circuits (Bravyi et al. 2021).

The benefit is not guaranteed

Beneficial on average but strongly platform-dependent (Russo et al. 2023).



Illustrative values (not project data).

Mitigation buys accuracy with extra quantum resources, so whether to use it is a real decision.

Source: Temme 2017 · Kandala 2019 · Giurgica-Tiron 2020 · Bravyi 2021 (PRA 103:042605) · Russo 2023 (IEEE TQE 4). Curve illustrative.

Proposed approach: Q-TRACE

QML model
on a benchmark task

→

Noisy execution
simulated with Qiskit Aer

→

Q-TRACE decides
benefit − cost > 0 ?

→

Mitigate or not
ZNE / readout mitigation

Benchmarks named in the record

Iris
small classification benchmark

WDBC
breast-cancer classification

QM9
molecular-property regression

Compared with three simple policies: never mitigate, always mitigate, and a fixed rule.

Recorded in docs/DECISIONS.md (Q1) and the early-October research plan's description of the deck of that time. The fraud application came later, on Oct 6, 2026.

Research questions and planned method

RQ1

When does error mitigation improve a quantum model's output enough to justify its extra cost?

RQ2

Can Q-TRACE's benefit-versus-cost decision beat never, always, or a fixed rule?

1. Simulate
noisy QML models (Qiskit Aer)

→

2. Measure
benefit and cost of mitigating vs not

→

3. Decide
weigh benefit against cost

→

4. Compare
vs never / always / fixed rule

Note added in October 2026. On Oct 6 the scope was narrowed to credit-card fraud detection, with several mitigation actions, label-free decision inputs and pre-registered phases (docs/DECISIONS.md: Q1, A01, A12). Later meetings report that study.

A practical question: spend quantum resources on mitigation only when it pays.

Source: Reconstructed from docs/DECISIONS.md (Q1) and docs/research/deep-research-report.md (pre-October deck); wording is ours.